From RANS Prediction to Hydrofoil Optimization: A Benchmark of Neural Surrogates

Anonymous Author(s)
Affiliation
Address
email

Abstract

Neural flow surrogates are usually compared by field error, although their practical value often lies in the designs they select. We compare CNN-U-Net, a physics-regularized point network, FNO, and DeepONet on 381 OpenFOAM simulations of hydrofoils in water. Complete families form disjoint training, validation, and final-test sets, and every architecture is trained with three seeds. The models predict RANS fields, lift, drag, and pressure-based cavitation-inception risk, and then drive the same four-parameter hydrofoil shape-optimization loop. DeepONet gives the lowest mean pressure, C_p, lift, and drag errors on the untouched test families; the physics-regularized model is close and fastest. Inference is 384–867 times faster than OpenFOAM. Across seeds, endpoint solves give mean L/D from 22.10 to 37.90, but a three-level grid study shows that drag remains mesh-sensitive. Field accuracy, force accuracy, search quality, and simulator verification are related but not interchangeable.

1 Introduction

For a hydrofoil operating in water, shape design is an optimization problem wrapped around a flow solver. Each change in camber, thickness, or incidence requires another RANS solution, so even a modest search can consume hundreds of CFD evaluations; neural surrogates move most of that search off the solver by learning the map from geometry and operating conditions to flow and forces [Brunton et al., 2020]. The difficult question is not simply which network minimizes field error, but which one preserves the quantities that determine a useful design.

CNN-U-Nets emphasize local multiscale structure [Ronneberger et al., 2015]; FNO mixes information globally in Fourier space [Li et al., 2021, Kovachki et al., 2023]; DeepONet separates parameter and coordinate representations [Lu et al., 2021]; and physics-informed models constrain learning with differential equations [Raissi et al., 2019]. Their relative merits may therefore change as the target moves from pressure fields to forces and finally to shape optimization.

Airfoil-surrogate research has progressed from field prediction [Guo et al., 2016, Bhatnagar et al., 2019, Sekar et al., 2019, Thuerey et al., 2020] to benchmarks that evaluate surface forces, out-of-distribution accuracy, and computational cost [Bonnet et al., 2022, Li et al., 2023, Yagoubi et al., 2025]. Those metrics establish whether a model approximates CFD, but they do not by themselves establish whether it will make the same design decision. A small, structured force error can be amplified when an optimizer repeatedly queries a narrow region of design space. The missing experiment is therefore a controlled comparison that follows several surrogate families through the same optimizer and then checks the

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

selected geometries with new CFD solutions. High-fidelity endpoint verification is standard in surrogate shape optimization, while a matched direct-CFD search provides a stronger reference when its cost is manageable [Li et al., 2022, Shukla et al., 2023].

That experiment is the contribution of this paper; we do not propose a new architecture. We train four representative model families on the same 381 RANS simulations and reserve two complete NACA families for validation. We compare fields, total forces, pressure-threshold cavitation screening, and runtime, then give every model the same three-start hydrodynamic shape optimization (HSO) problem. Fresh OpenFOAM solutions check the four surrogate winners, while a direct OpenFOAM search uses the same starts and budget as a reference. Cavitation risk here means inception inferred from C_p, not a multiphase prediction of cavity evolution.

2 Data and CFD Reference

2.1 Design space

The dataset contains NACA 0006, 0008, 0010, 0012, 0015, 0018, 2412, 2415, 4412, 4415, 4418, and 4421 profiles generated from the standard four-digit parameterization Abbott and von Doenhoff [1959]. For each profile, we simulate angles of attack $\alpha \in \{-4, -2, 0, 2, 4, 6, 8, 10\}^\circ$ and Reynolds numbers $Re \in \{10^5, 2 \times 10^5, 5 \times 10^5, 10^6\}$. All cases use liquid water with density 997 kg/m^3 and kinematic viscosity 10^{-6} m^2/s. With a 1 m chord, the freestream speed is set by $U_\infty = Re\nu/c$; the far-field absolute pressure and vapor pressure are 101325 and 2300 Pa.

Each case is solved with OpenFOAM `simpleFoam` [Weller et al., 1998]. For steady incompressible flow, the mean velocity $\bar{u}_i$ and pressure $\bar{p}$ satisfy

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0, \tag{1}$$

$$\rho \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \mu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \rho \frac{\partial \overline{u_i' u_j'}}{\partial x_j}. \tag{2}$$

The Reynolds stress is closed with the k–ω SST model [Menter, 1994]. Angled freestream velocity is imposed at the inlet and far boundary, and OpenFOAM's `forceCoeffs` supplies total lift and drag after 300 iterations. Cell-centered fields are interpolated to a common 128×64 Cartesian grid over $x/c \in [-1, 2]$ and $y/c \in [-0.75, 0.75]$. Inputs include coordinates, operating parameters, three NACA descriptors, a fluid mask, and signed distance to the profile.

Three simulations with $|C_L|$ or $|C_D| > 5$ are discarded before splitting, leaving 381 cases. NACA 0015 and 2412 provide 64 validation cases for early stopping; NACA 0018 and 4418 provide an untouched 64-case test set used only for final reporting. The remaining 253 cases from eight families train the models. This fixed partition is identical for every architecture and seed, and no validation or test geometry appears in training at another operating point. All retained solves reach iteration 300. The 95th-percentile maximum initial residual at that iteration is 1.27×10^{-4}; the corresponding late-window spans are 0.0059 in C_L and 2.65×10^{-4} in C_D.

2.2 Targets and cavitation-risk definition

The common output vector contains U_x, U_y, absolute pressure p, eddy viscosity, k, ω, C_p, cavitation margin, C_L, and C_D. Grid fields use masked normalized mean-squared error, while the case-level force targets use normalized mean-squared error. OpenFOAM's kinematic gauge pressure p_k is converted to absolute pressure as

$$p = p_\infty + \rho p_k, \qquad C_p = \frac{p - p_\infty}{\frac{1}{2}\rho U_\infty^2}. \tag{3}$$

For an ambient-pressure value p_a, the pressure-based inception margin is

$$M(x, y; p_a) = p_a + C_p(x, y)\tfrac{1}{2}\rho U_\infty^2 - p_v. \tag{4}$$

A cell is labeled at risk when $M < 0$. We evaluate this label at $p_a \in \{2500, 3000, 5000, 10000, 25000, 101325\}$ Pa. This sweep creates a useful classification test from the predicted C_p field without representing multiphase physics.

3 Surrogates and Optimization

3.1 Model families

The CNN-U-Net has three encoder/decoder levels, skip connections, group normalization, GELU activations, and base width 16. The FNO has width 32, four operator blocks, and 12 retained Fourier modes per direction Ronneberger et al. [2015], Li et al. [2021]:

$$h_{l+1} = \sigma\big(W_l h_l + \mathcal{F}^{-1}(R_l \mathcal{F}(h_l))\big) . \tag{5}$$

The DeepONet uses three-layer, width-64 branch and trunk networks with 64 basis functions. Its output has the form

$$\widehat{y}_k(\mathbf{x}; \boldsymbol{\mu}) = \mathbf{b}_k(\boldsymbol{\mu})^\top \mathbf{t}_k(\mathbf{x})/\sqrt{64} + c_k. \tag{6}$$

following Lu et al. [2021]. The physics-regularized point network has three tanh layers of width 64 and uses 1,024 sampled points per case. Its supervised loss is augmented by 10^{-4} times the continuity and first-order momentum residual loss Raissi et al. [2019]. This is a lightweight physics regularizer, not a data-free PINN or complete RANS residual.

All models use AdamW with learning rate 2×10^{-3}, weight decay 10^{-4}, at most 120 epochs, patience 20, and gradient clipping. Each architecture is trained with seeds 7, 17, and 27 on the same split. Normalization uses training cases only. The best checkpoint is selected by validation loss. Model size ranges from 9,930 to 595,914 parameters (Table 1).

Data, targets, optimizer, stopping rule, and evaluation budget are shared, but parameter counts are reported rather than artificially matched: equal width is not equal capacity across convolutional, spectral, branch–trunk, and pointwise models. These are compact canonical configurations, and no hyperparameter is selected on the test families.

Optimization uses dedicated C_L and C_D output heads supervised by OpenFOAM's total force coefficients. Pressure integration on the coarse common grid is not used for the objective because it omits viscous traction and under-resolves the wall.

3.2 Common optimization protocol

At $Re = 5 \times 10^5$, each surrogate drives the same HSO loop. The design vector $\theta = (m, p, t, \alpha)$ contains maximum camber, camber location, thickness, and incidence, with bounds $m \in [0, 0.09]$, $p \in [0.1, 0.9]$, $t \in [0.06, 0.20]$, and $\alpha \in [-4, 10]^\circ$. We launch Nelder–Mead [Nelder and Mead, 1965] from NACA 0012, 2412, and 4415 and give every run the same 40-iteration budget. The objective is

$$J(\theta) = \frac{C_L}{|C_D| + 10^{-6}}. \tag{7}$$

The search calls only the surrogate; OpenFOAM is not used inside the loop, and cavitation margin is retained only as a secondary diagnostic. We return the best design observed within the fixed budget, select the strongest of the three starts for each architecture, and then generate a new mesh and independent OpenFOAM simulation for that winner.

The optimizer is continuous although the training geometries are discrete. It queries intermediate four-digit descriptors (m, p, t) within the stated bounds and can move beyond the convex coverage of the training families near a bound. We retain every query and independently remesh and solve each selected endpoint with CFD; this validates the reported design, not the full learned surface.

The direct-CFD baseline repeats this protocol with the same starts, bounds, objective, tolerances, and 40-iteration budget, but OpenFOAM supplies every objective value. It is distinct from the one-solve endpoint checks of the surrogate winners.

Table 1: Final-test RMSE (mean ± sample standard deviation over three seeds), size, and speedup.

Model	p (Pa)	C_p	C_L	C_D	Params.	Speedup
CNN-U-Net	28.32 ± 14.18	$0.0657 \pm .0373$	$0.258 \pm .200$	$0.00871 \pm .00522$	485,178	544 ± 13
Physics-reg. MLP	11.56 ± 0.48	$0.0445 \pm .0012$	$0.0352 \pm .0056$	$0.00328 \pm .00029$	9,930	867 ± 1
FNO	27.79 ± 1.07	$0.0643 \pm .0040$	$0.248 \pm .020$	$0.0103 \pm .00054$	595,914	464 ± 6
DeepONet	10.74 ± 1.48	$0.0411 \pm .0045$	$0.0304 \pm .0081$	$0.00339 \pm .00089$	100,746	384 ± 2

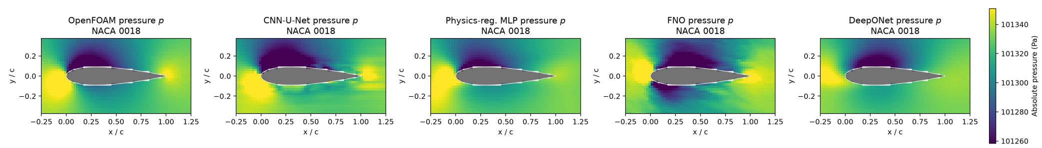

Figure 1: Absolute-pressure prediction for a representative NACA 0018 test case (case 179), shown over $x/c \in [-0.25, 1.25]$ and $y/c \in [-0.375, 0.375]$. Every panel uses the same color scale.

4 Results

4.1 Final-test accuracy and computational cost

Table 1 reports mean and one sample standard deviation over three seeds on the untouched NACA 0018/4418 test set. DeepONet gives the lowest mean pressure, C_p, C_L, and C_D errors. The physics-regularized model is consistently second; CNN-U-Net is notably seed-sensitive.

Mean training times are 556, 67, 270, and 47 s for CNN-U-Net, the physics-regularized network, FNO, and DeepONet. Median OpenFOAM execution time is 12.4 s per case, while complete-field inference takes 14–32 ms, yielding 384–867× speedups. The benchmark used an 8-core Apple M2 Mac mini with 16 GB memory; PyTorch ran on CPU and OpenFOAM 9 ran in Docker.

Figure 1 shows a representative final-test case from the unseen NACA 0018 family. The pointwise models reproduce the leading-edge pressure structure most closely; CNN-U-Net smooths the extrema and FNO introduces spatial texture. A common color scale makes those differences directly comparable across all five panels.

4.2 Cavitation-inception screening

The pressure-threshold labels are highly imbalanced, so we report F1 rather than accuracy. DeepONet obtains 0.900 ± 0.008, the physics-regularized model 0.867 ± 0.008, CNN-U-Net 0.758 ± 0.198, and FNO 0.724 ± 0.065. This is a secondary low-C_p diagnostic, not a prediction of cavity dynamics Brennen [1995].

4.3 Shape optimization against direct CFD

Figure 2 reports the complete L/D comparison: the OpenFOAM value of the starting foil, the surrogate estimate at its selected design, and the result of remeshing that design and solving it once in OpenFOAM. This one-solve endpoint check is distinct from the direct baseline, which places OpenFOAM inside the matched optimizer.

Across three selected endpoints per architecture, medium-grid OpenFOAM gives $L/D = 24.44 \pm 1.35$ for CNN-U-Net, 37.90 ± 1.59 for the physics-regularized network, 22.10 ± 3.46 for FNO, and 37.47 ± 5.95 for DeepONet. The matched direct-CFD search reaches 29.89. These values are not mesh-independent: on the seed-7 winners, medium-to-fine L/D shifts are 16–20%.

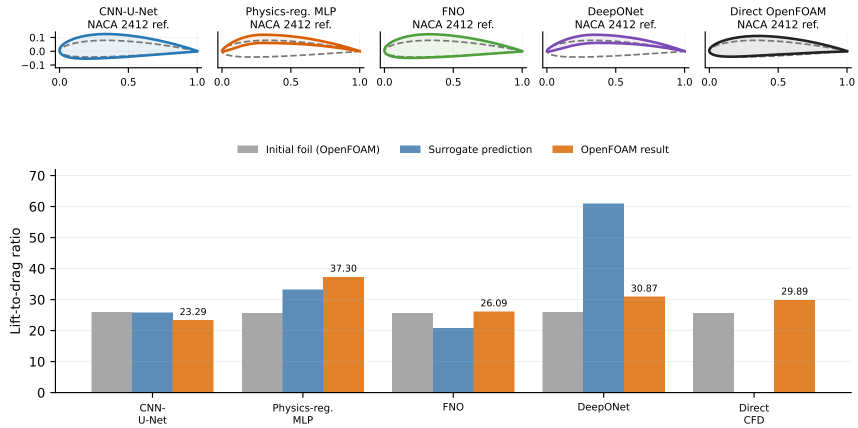

Figure 2: Top: selected shapes against the same dashed NACA 2412 geometric reference. The reference is for visual comparison, not necessarily the optimizer initializer. Bottom: initial-foil OpenFOAM values, surrogate predictions, fresh OpenFOAM endpoint checks, and direct CFD.

5 Discussion

DeepONet is the best final-test predictor, but its optimized endpoints vary more across seeds than those of the physics-regularized network. CNN-U-Net is unstable in both force prediction and optimization; FNO is more repeatable but selects lower-performing designs. Accuracy, candidate ranking, and seed robustness are therefore distinct.

All 12 grid-study runs converge iteratively by 1,000 iterations, with final residuals below 10^{-7} and negligible late force drift. Lift changes by at most about 4% from medium to fine, but drag changes by 16–24%. Thus, a fresh OpenFOAM endpoint is a standard and useful verification step, yet it is not by itself a grid-independent validation. This is precisely the simulator imperfection exposed by carrying the benchmark through the design loop.

The continuous optimizer also queries descriptors between and near the bounds of twelve discrete families. Endpoint CFD checks the selected shape, not the entire learned surface. All winners remain far from vapor pressure, so cavitation is a secondary diagnostic rather than an active constraint. The remaining scope is steady 2D RANS and the NACA four-digit space.

6 Conclusion

On an untouched family-level test set, DeepONet gives the lowest mean field and force errors, while the smaller physics-regularized model is close and fastest. Following all models through a multi-start optimizer reveals seed-dependent design decisions that test error alone does not predict. The endpoint and grid checks add the crucial final result: a surrogate should be judged not only by prediction and speed, but by the consequence of acting on it and the numerical uncertainty of the simulator used to verify that action.

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A Complete Optimization Traces

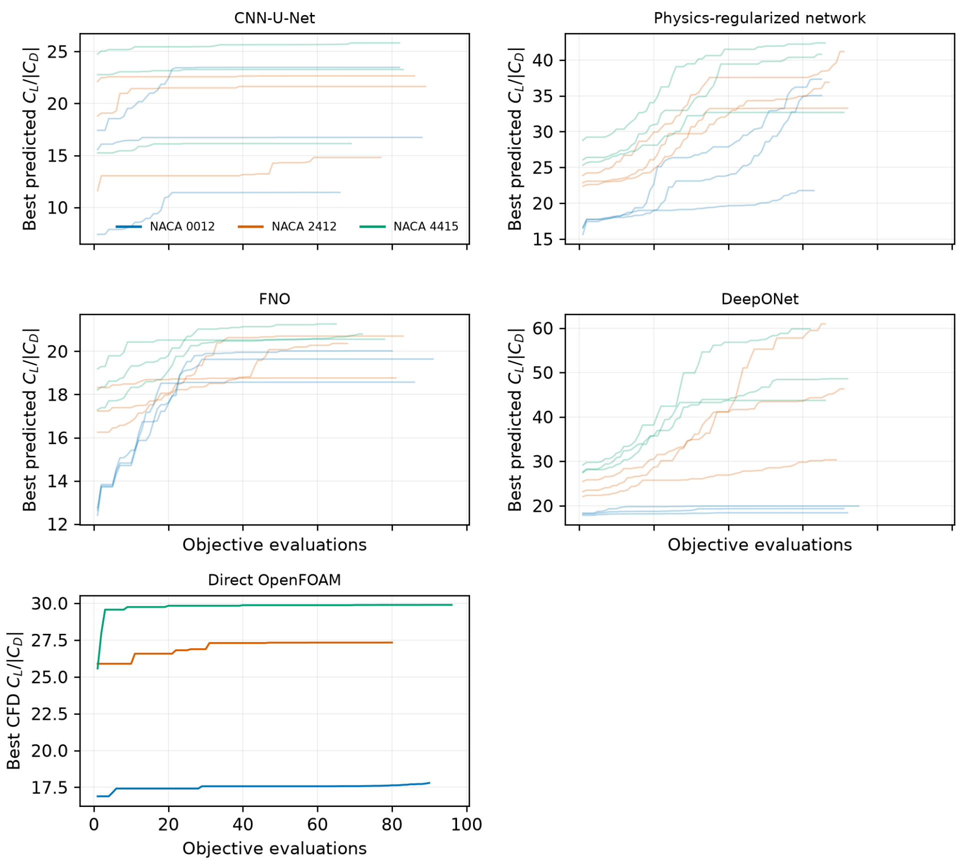

Figure 3: All three optimization starts and three training seeds per surrogate, with the three direct-CFD starts. Lines show best-so-far objective; every raw evaluation is included in the anonymous artifact.